

Which physicians should 60 reps call on?

- Northwind Pharma markets a branded DOAC with a fixed 60-rep field force,
- roughly \$15M a year fully loaded, and flat share for two years.
- Built on 1,380,665 Medicare Part D prescribers, 2022-2024.

SITUATION

The industry ranks doctors by prescription volume

- That rule is structurally wrong.
- A prescriber already at 90% brand share has no headroom to win.
- A mid-volume prescriber at 6% share is where the growth is.
- Volume ranking cannot tell them apart.

Model what each prescriber could reach, then subtract

- Quantile-regression frontier at $\tau = 0.80$ on practice and market covariates.
- $\text{opportunity} = \text{potential_class} \times \text{achievable_share} - \text{brand_fills}$
- Frontier coverage 0.785; SFA cross-check 0.8494.
- No brand-volume feature reaches the potential model - the leakage guard raises.

29 GB of source, streamed and reduced to 200 MB

- 13.7% of all claim volume is suppressed by CMS,
- rising to 68.7% for prescribers under 100 claims - and it is not random.
- Rows under 11 claims are REMOVED, not blanked; the gap is recovered by
- reconciling provider totals against the sum of drug rows.

Volume and opportunity disagree for 59.4% of prescribers

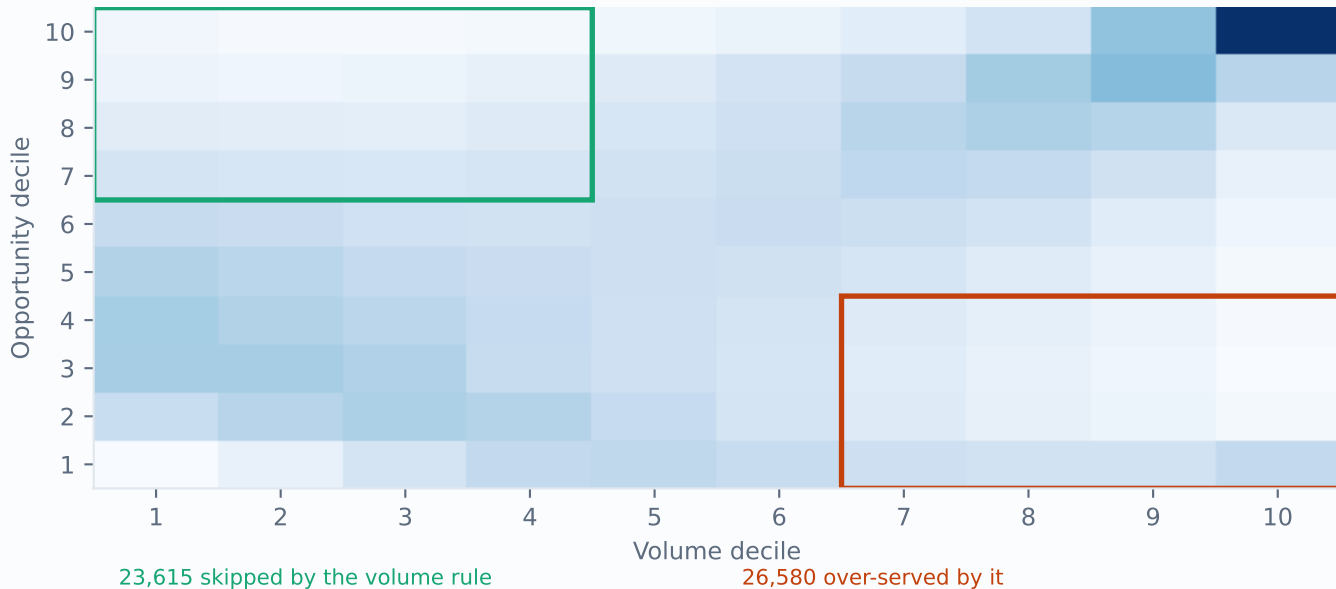


EXHIBIT E2 / BACK-TESTED

Held-out year: the ranking ranks (Spearman 1.0)

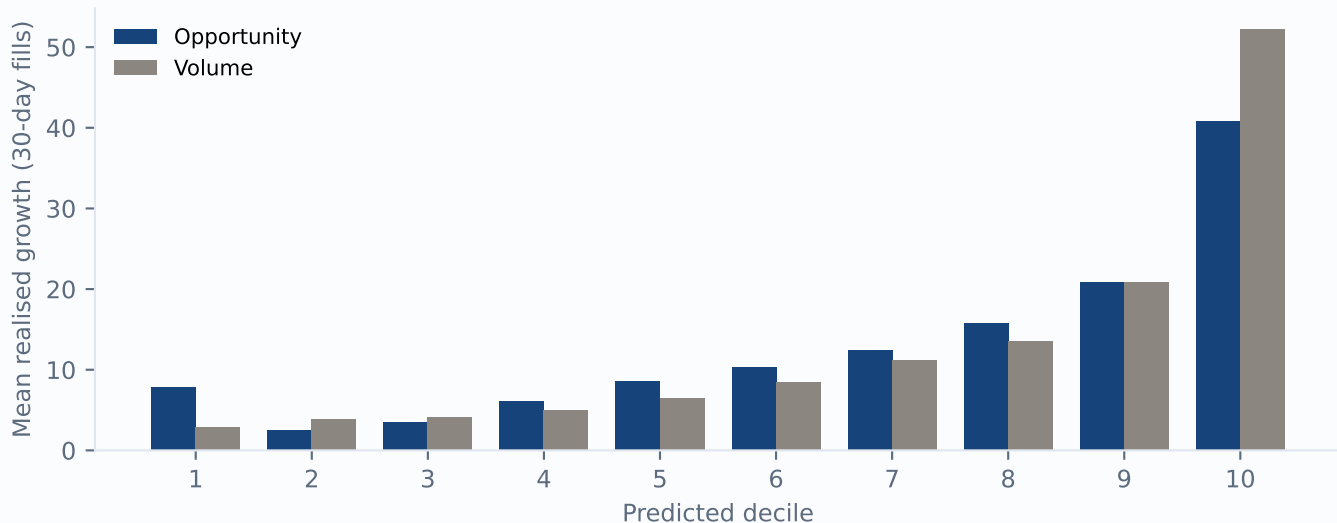
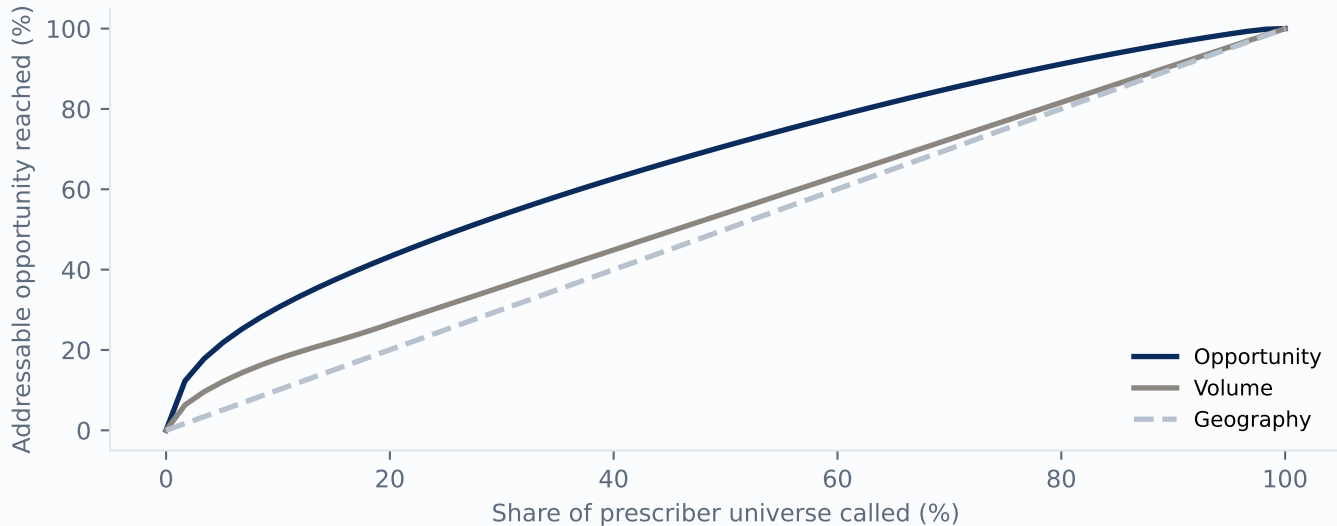
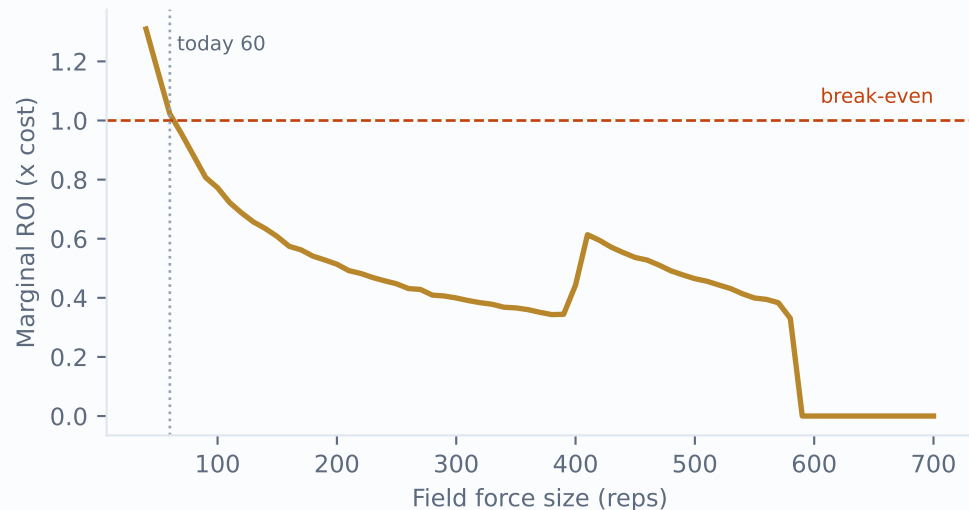
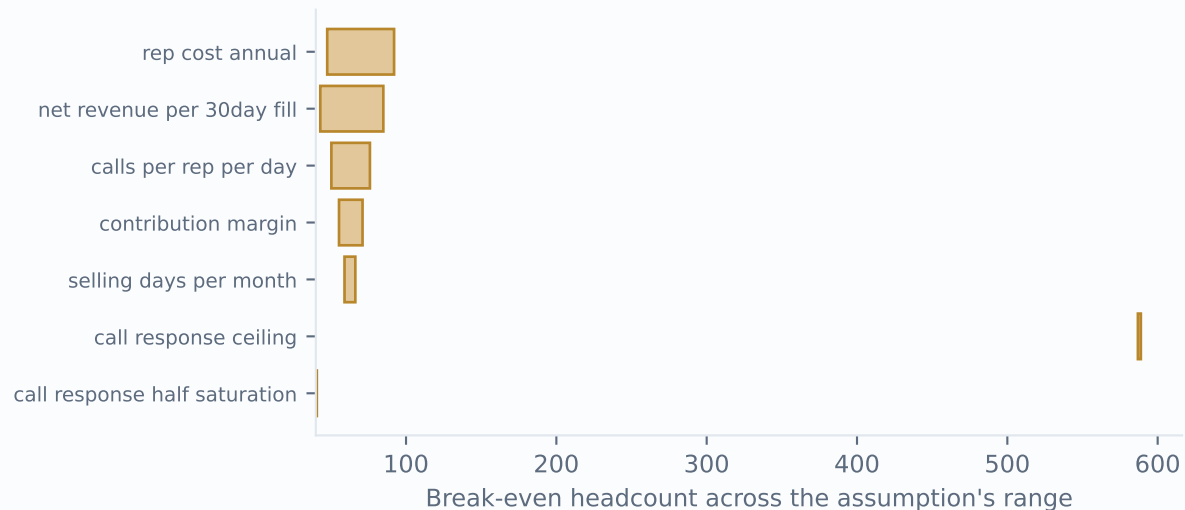


EXHIBIT E3 / ARITHMETIC

Same budget, 6.9% reached vs 3.2%

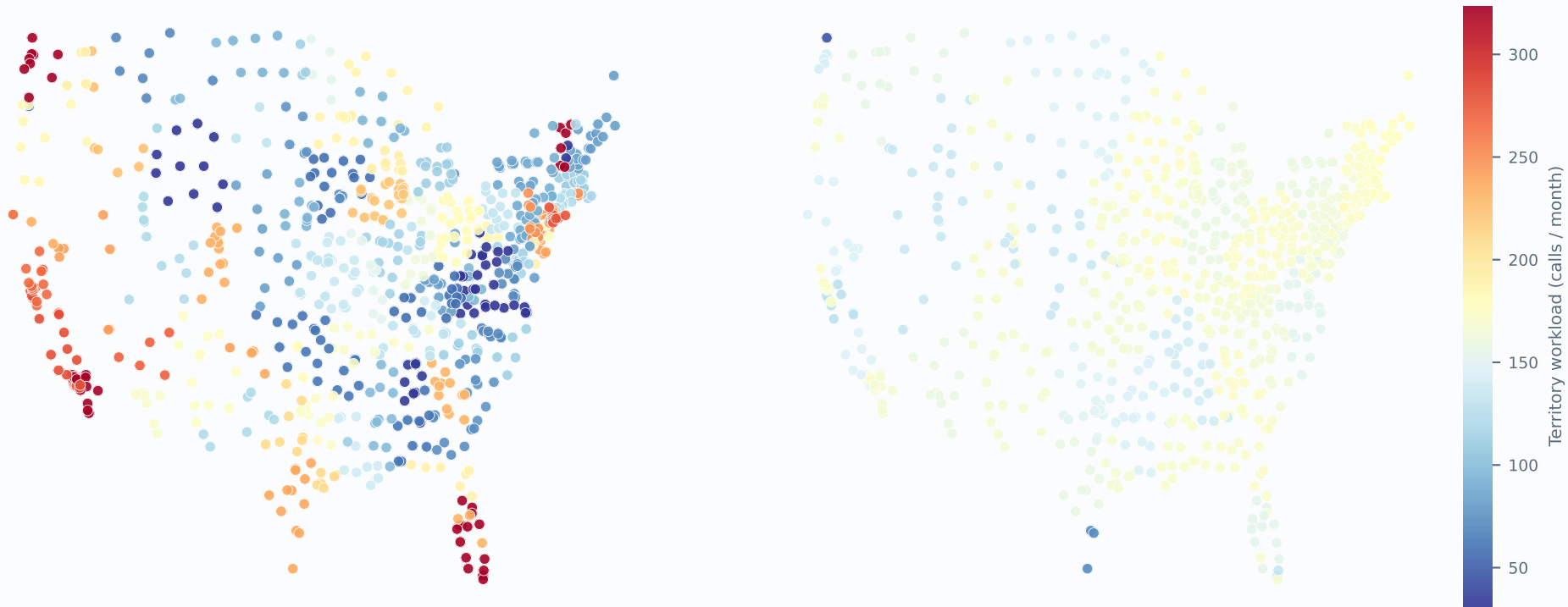


Marginal-rep economics**Which assumption to argue with first**

Territory alignment: imbalance 157x -> 2.5x | contiguity 20% -> 100% | travel -75%

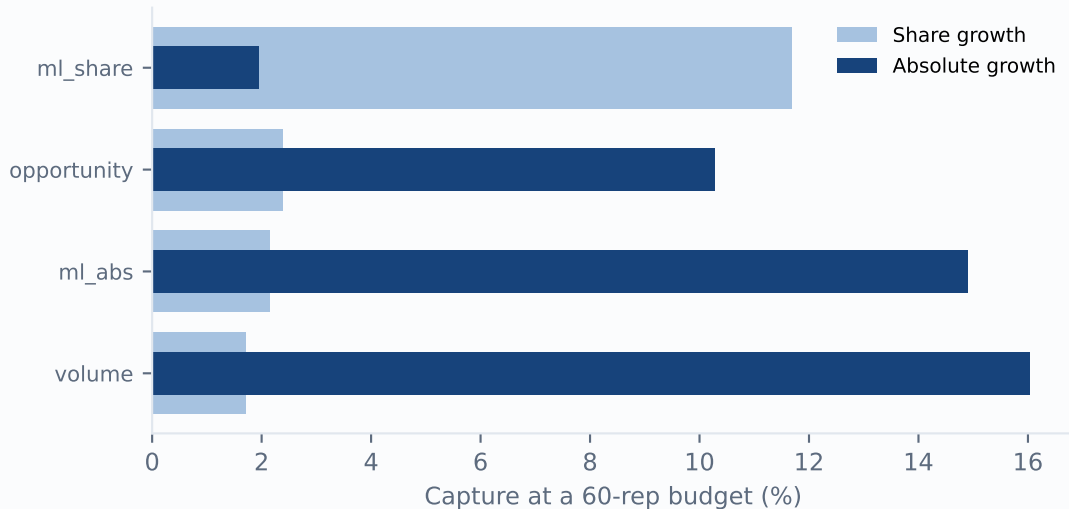
Before - alphabetical by state

After - capacity-constrained, contiguous



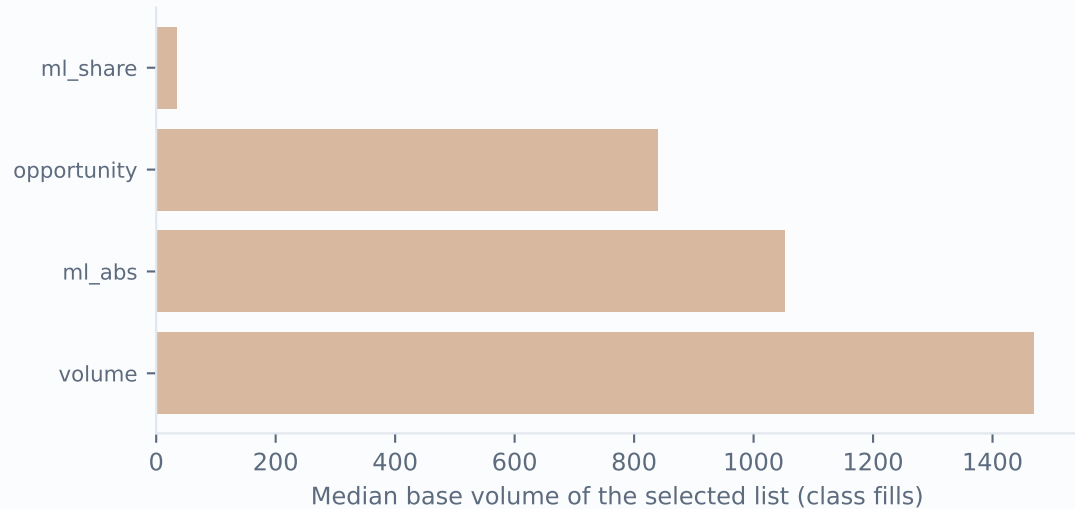
CHAMPION VS CHALLENGER

The ML model wins the stated metric



THE DIAGNOSTIC

...by selecting micro-prescribers



FINDING 1

The two rankings disagree for 59.4% of prescribers

- 23,615 are invisible to the volume rule.
- 26,580 are over-served by it.
- That disagreement is the entire argument for the project.

FINDING 2

Flagged prescribers grew 0.507x faster in a held-out year

- Spearman 1.0 between predicted decile and realised growth.
- Share-growth capture 1.4x the volume rule.
- Absolute-growth capture 0.8x - volume wins this one.
- (Absolute growth scales with baseline volume, so volume is mechanically advantaged.)

FINDING 3

The field force is drastically under-sized

- 60 reps serve 8,389 of 267,171 in-market prescribers - 3.1% - at full capacity.
- The unconstrained call plan wants 106,869 prescribers.
- (Headcount economics are a scenario; the coverage gap is arithmetic.)

RECOMMENDATION

Re-rank on opportunity; hold headcount; redraw the map

- Reach rises from 3.2% to 6.9% of addressable opportunity, same budget.
- Territory imbalance falls 157x -> 2.5x with 100% contiguity and 75% less travel.
- Neither requires a single additional representative.

What this cannot tell you

- Medicare-only; ~17-month lag; suppression hides small prescribers.
- Promotional payments: parallel trends FAILED, so association only.
- No call data, so the back-test sees where growth happened,
- not where a call would have caused it. Volume ranking wins absolute capture.
- (An ML challenger beat the stated metric by selecting 35-fill prescribers.
- The metric was gameable; the gate was hardened rather than the result kept.)

NEXT

What a brand team would do with this

- Replace Part D with all-payer IQVIA to remove the Medicare skew.
- Join rep call logs to make the response curve empirical rather than assumed.
- Add drive-time matrices and org boundaries before executing any alignment.
- Re-run quarterly: the pipeline is resumable and takes ~20 minutes end to end.